

Innovation and Isolation: Status Rents and Endogenous Networks

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Abstract

I study how groups form innovation networks when power determines their claims on resources. Cooperation creates resources but can also threaten those claims. Under the benchmark rule for changes in power, there is a greatest equilibrium: a network containing every other equilibrium network. Every active link operates at its maximum feasible strength. A group that loses from innovation cuts all its links, whatever its partners' rank. Complementarities transmit withdrawal: losing a partner can make links among the remaining groups unprofitable even when their private values of innovation stay positive. I characterize exactly when withdrawal induces these additional losses and, when status exposure rises, separate lost complementary returns from changed incentives. When power and link capacity are aligned, vetoes by top-ranked groups remove the strongest feasible links. The veto requires that a group expect to lose share in a redraw, not that the redraw equalize shares, and it survives nonlinear status lotteries under a marginal condition preserved by sufficiently small admissible smooth perturbations of a strict benchmark veto. Stronger spillovers or larger innovation inputs weakly expand the greatest equilibrium, while greater status exposure can first increase interaction and then reduce it. With any number of groups, full integration requires exposure low enough to retain advantaged groups and high enough to attract disadvantaged groups. Full integration, when feasible, is sustained over a single interval of exposure. Even reimbursing all interaction costs cannot secure a group's consent if innovation reduces its expected consumption. Compensating that loss requires transfers enforceable after power changes, which the model assumes are unavailable.

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1 Introduction

Innovation creates resources, but it can also change who gets them. Groups that receive a large share may therefore avoid the interactions that help society innovate. This paper asks when protecting that share leads groups to reject useful cooperation. Their withdrawal can be especially costly when they are also the most productive partners.

If cooperation creates a surplus, its beneficiaries might compensate groups that would otherwise oppose it. [Acemoglu \(2025\)](#) distinguishes economic losses, which transfers could offset if adequate fiscal instruments exist, from political losses, which can deprive a group of the power to secure compensation. [Acemoglu \(2003\)](#) explains why this commitment problem can prevent mutually beneficial political bargains. Institutions that constrain future rulers can make promises credible ([North and Weingast, 1989](#)). Here, claims on resources follow political priority, and no agreement guarantees compensating payments after priority changes. Under this assumption, the threat of losing power can prevent consent to productive links.

Groups choose whom to interact with and how strongly. Each link, an edge in a graph whose nodes are the groups, requires both groups to agree. Innovation spreads through the network and increases total resources. Groups receive these resources in order of priority, and more innovation makes a random change in that order more likely. Power determines both a group’s initial claim on resources and how strong its links can be. Historical evidence links political power to unequal access to resources. [Sokoloff and Engerman \(2000\)](#) describe how elites in the Americas shaped land, voting, and schooling policies to preserve their advantages. In Colombia, historical political concentration is associated with worse later development ([Acemoglu et al., 2008](#)), and [Robinson \(2016\)](#) argues that extractive political institutions help sustain poverty and violence. [Crabtree and Durand \(2017\)](#) examine how business elites influence the Peruvian state. Other studies document the lasting economic costs of forced mining labor in Peru ([Dell, 2010](#)) and paramilitary influence on Colombian elections ([Acemoglu et al., 2013](#)). On technology adoption, [Comin and Hobijn \(2009\)](#) find cross-country evidence consistent with barriers erected by incumbent lobbies. The model focuses on a specific decision: whether a group consents to interaction. The threat to its claim comes from aggregate innovation, whoever produces it, so a group that loses from innovation cuts all its links, to higher-ranked and lower-ranked partners alike. The model therefore explains disconnection, the loss of productive interaction, and not which groups are kept apart from which.

The benchmark makes the probability of a new priority draw proportional to innovation divided by total resources. This functional form is an assumption. It represents the possibility that economic change weakens incumbents: [Acemoglu and Robinson \(2006\)](#) model technological change that raises the risk of political replacement, while [Acemoglu et al. \(2005\)](#) show how Atlantic trade strengthened merchant groups where initial institutions limited royal power. With linear utility, expected consumption becomes affine in innovation, which permits a global equilibrium characterization. The isolation mechanism needs less: under a general increasing lottery, a group withdraws when the marginal loss of status rents exceeds its productive gain. A strict veto survives small smooth perturbations of the benchmark and a nonlinear alternative lottery. Nor does the veto rest on a redraw that equalizes expected shares. Any redraw after which a group expects a smaller share than it holds produces a veto once exposure reaches a threshold set by the size of that loss, provided the threshold is admissible. The ordering of these thresholds by rank holds under equalization but not in general. Reimbursing link costs cannot remove a veto caused by lower expected consumption; compensation must cover that loss itself.

The economic role of the network is to transmit one group’s withdrawal to cooperation among other groups. Links complement one another in spreading innovation. Losing a partner can therefore

destroy the incentive to maintain links that do not involve that partner, even if the remaining groups' private innovation values are unchanged and positive. Starting from the greatest equilibrium, I give a necessary and sufficient condition for these additional losses: the reduction in complementary returns must exceed the retention margin for some set of links. When exposure itself rises, the remaining groups' values change along with the network. An exact identity then splits the change in each retention test into the complementary returns lost with the vetoing groups' links and the change in the group's own value. A group whose value has not fallen can fail a retention test only because of lost complementary returns. A three-group example produces complete isolation although the remaining groups still want more innovation. This propagation is absent when innovation is additive across links. The ordering of capacities by power has a separate implication: when status concerns isolate the highest-ranked groups and at least two groups remain outside those ranks, the share of feasible capacity lost exceeds the share of pairs removed.

In equilibrium, every active link operates at its maximum feasible strength, and no group gains by dropping any set of its links. There is an equilibrium network containing all the links present in any other equilibrium. Holding status exposure fixed, lower costs, stronger spillovers, or larger innovation inputs preserve its links and may add new ones. Greater exposure gives disadvantaged groups more reason to cooperate and advantaged groups more reason to withdraw. Interaction can first rise and then fall, as the two-group solution and four-group example show. For any number of groups, full integration requires exposure high enough for disadvantaged groups to maintain all their links and low enough for advantaged groups to do so. The corollary gives the exact interval where all groups are willing to maintain their links.

The paper studies network formation when innovation threatens existing claims on resources. It combines allocation by power (Piccione and Rubinstein, 2007) with the threat of political replacement (Acemoglu and Robinson, 2006). A related conflict appears in Dasaratha (2023), where firms weigh the benefits of learning from others against the risk that competitors learn their ideas. Here, groups may refuse productive links because the innovation they generate can reduce their share of total resources. Their withdrawal changes the pattern of social interaction, which Echenique and Fryer (2007) place at the center of segregation measurement. Their index depends on contacts within the same group and on how segregated those contacts are. With each group represented by a single node, the present model instead tracks how much interaction between groups is lost: the isolation index S is the share of feasible interaction left unused.

The network framework draws on bilateral consent (Jackson and Wolinsky, 1996) and the choice of link strength (Bloch and Dutta, 2009; Baumann, 2021). Its spillover structure is related to the complementarities studied on a given network by Ballester et al. (2006). The equilibrium analysis relates to Hellmann (2013) on the existence of stable networks and to Acemoglu and Tahbaz-Salehi (2025), who characterize a largest equilibrium production network and its response to costs and productivity. In the present setting, these tools show how the protection of resource shares shapes cooperation and how one group's withdrawal can lead others to sever their links.

Section 2 presents the model. Section 3 describes the networks that can arise in equilibrium. Section 4 studies how cooperation responds to costs, productivity, and the risk of changes in power. Section 5 examines welfare and compensation, with a four-group example. Section 6 considers alternative rules for changes in power. Section 7 concludes and discusses questions for future research.

2 Environment

2.1 Interaction technology and choices

There are $N \geq 2$ equally weighted groups, $V = \{1, \dots, N\}$, indexed by initial priority, $1 \succ_0 \dots \succ_0 N$. Each unordered pair ij has feasible intensities

$$\mathcal{R}_{ij} = \{0\} \cup [\underline{\rho}_{ij}, \bar{\rho}_{ij}], \quad 0 < \underline{\rho}_{ij} \leq \bar{\rho}_{ij}. \quad (1)$$

A pair either does not interact or interacts at an intensity between $\underline{\rho}_{ij}$, the smallest active intensity, and $\bar{\rho}_{ij}$, the pair's capacity. To link capacity to power, set $t(i, j) = \min\{i, j\}$ and $(\underline{\rho}_{ij}, \bar{\rho}_{ij}) = (\underline{\rho}_{t(i, j)}, \bar{\rho}_{t(i, j)})$, where $\bar{\rho}_{r+1} < \underline{\rho}_r$ for $r < N - 1$. Hence, the more powerful endpoint determines the pair's productive class. This represents access to infrastructure, organizational resources, or knowledge that a powerful partner can bring to cooperation. Equation (24) identifies the resulting concentration of lost capacity when high-ranked groups withdraw. Cross-status links remain feasible because both endpoints face the same pair-specific band.

Group i offers $\ell_{ij} \in \mathcal{R}_{ij}$ to each other group $j \neq i$. Realized intensity is $\rho_{ij} = \min\{\ell_{ij}, \ell_{ji}\} = \rho_{ji}$, with $\rho_{ii} = 0$. Either group can reduce or end interaction by lowering its offer. Interaction at intensity t requires both groups to offer at least t . Let $\bar{\rho} = \max_{i < j} \bar{\rho}_{ij}$ be the largest capacity across all distinct pairs and define

$$\mathbf{G}(\boldsymbol{\rho}) = [\rho_{ij}/\bar{\rho}]_{i,j=1}^N, \quad K_i(\boldsymbol{\rho}) = \sum_{j \neq i} (\kappa_0 \mathbf{1}\{\rho_{ij} > 0\} + \kappa_1 \rho_{ij}), \quad \kappa_0 > 0, \quad \kappa_1 \geq 0. \quad (2)$$

The matrix $\mathbf{G}$ records link strengths on a common scale from zero to one. Dividing every intensity by the same $\bar{\rho}$ preserves differences in strength across links. In K_i , each active link costs group i a fixed amount κ_0 plus $\kappa_1 \rho_{ij}$, which rises with intensity when $\kappa_1 > 0$. Each group pays its own costs, measured in utility units. Costs represent effort rather than an additional subtraction from the resource stock. Unaccepted offers are costless. The scale $\bar{\rho}$ stays fixed when capacities are held fixed.

Integration and isolation are, respectively,

$$C(\boldsymbol{\rho}) = \frac{\sum_{i < j} \rho_{ij}}{\sum_{i < j} \bar{\rho}_{ij}}, \quad S(\boldsymbol{\rho}) = 1 - C(\boldsymbol{\rho}). \quad (3)$$

Thus S measures unused interaction between groups. Two networks can have the same total interaction but produce different amounts of innovation because they connect different groups. Nor does S record who is connected to whom. With $N \geq 3$, a network whose only link is pair 12 at capacity and one whose only link is pair 13 at capacity have the same S , although the first joins the two highest ranks and the second reaches a lower one.

2.2 Innovation and resources

After offers are chosen, a vector of innovation inputs $\boldsymbol{\xi} = (\xi_1, \dots, \xi_N)$ is drawn from a commonly known distribution independent of the offers, where $\xi_i \geq 0$ is the input of group i . Assume $\boldsymbol{\xi} \leq \bar{\mathbf{v}}$ almost surely for a fixed vector of upper bounds $\bar{\mathbf{v}}$, and let $\mathbb{E}[\boldsymbol{\xi}] = \mathbf{v} \gg \mathbf{0}$, so that the entry v_i of $\mathbf{v}$ is the expected input of group i . Let $\mathbf{y} = (y_1, \dots, y_N)$ collect realized innovation, where y_i is the innovation of group i . Conditional on the graph, innovation satisfies

$$\mathbf{y} = \boldsymbol{\xi} + \beta \mathbf{G}(\boldsymbol{\rho}) \mathbf{y}, \quad 0 < \beta < \frac{1}{N-1}. \quad (4)$$

Each group's innovation is its own input plus a fraction β of its partners' innovation, weighted by relative link intensity. The unique solution is $\mathbf{y} = \mathbf{M}(\boldsymbol{\rho})\boldsymbol{\xi}$, where

$$\mathbf{M}(\boldsymbol{\rho}) = [\mathbf{I} - \beta \mathbf{G}(\boldsymbol{\rho})]^{-1} = \sum_{k \geq 0} \beta^k \mathbf{G}(\boldsymbol{\rho})^k. \quad (5)$$

Every entry of $\mathbf{G}(\boldsymbol{\rho})$ lies in $[0, 1]$ and its diagonal is zero, so each row sum is at most $N - 1$. By Gershgorin's theorem (Horn and Johnson, 2013, Theorem 6.1.1), the spectral radius of $\beta \mathbf{G}(\boldsymbol{\rho})$ is at most $\beta(N - 1) < 1$ for every feasible network, so $\mathbf{I} - \beta \mathbf{G}(\boldsymbol{\rho})$ is invertible and the series converges. The (i, j) entry of $\mathbf{G}^k$ adds, over all walks of length k from i to j , the product of the normalized intensities along the walk. Hence y_i collects the input of every group j through every walk from i to j , discounted by β per step and weighted by the intensities along the walk. Walks may return to a group already visited, so innovation that flows out and comes back adds to it: cycles are productive. Define realized average innovation $z = N^{-1} \mathbf{1}^\top \mathbf{y}$ and its expectation

$$h(\boldsymbol{\rho}) = \mathbb{E}[z] = \frac{1}{N} \mathbf{1}^\top \mathbf{M}(\boldsymbol{\rho}) \mathbf{v}. \quad (6)$$

The resource stock is a scalar: the economy holds $Y(z) = Y_0 + Dz$ units of a common consumption basket, introduced below, where $Y_0 > 0$ is the baseline stock and $D > 0$ converts average innovation into resources. Innovation has no separate direct utility value.

2.3 Priority, consumption, and the status lottery

Each group consumes a scalar quantity q_i of a common basket $\mathbf{e} \neq \mathbf{0}$, so its bundle is $q_i \mathbf{e}$ and its utility is q_i . Feasible allocations satisfy $\mathbf{q} \in \mathbb{R}_+^N$ and $\sum_i q_i \leq Y$. Priority determines shares through a fixed profile. If $\sigma(r)$ occupies rank r , then

$$q_{\sigma(r)} = Y a_r, \quad a_1 \geq a_2 \geq \dots \geq a_N > 0, \quad \sum_{r=1}^N a_r = 1, \quad a_1 > \frac{1}{N}. \quad (7)$$

Here $\sigma(r)$ is the group that holds rank r after the status draw, and a_r is the fraction of the stock attached to that rank. Every group consumes, a higher rank carries a weakly larger fraction, and the profile is not flat, so the top rank consumes more than the average $1/N$. The profile summarizes power in the allocation, in the spirit of the jungle of Piccione and Rubinstein (2007), where stronger agents are served first. Group i starts at rank i , so its initial share is a_i . A uniform random permutation gives each group every rank with probability $1/N$, hence expected share $\sum_r a_r / N = 1/N$.

Example 1 (Allocation by priority). If each group can absorb at most a share $b \in (1/N, 1/(N - 1))$ and groups are served in order, the first $N - 1$ take b and the last takes the remainder $1 - (N - 1)b > 0$.

Let $\boldsymbol{\rho}^{\max}$ put every pair at capacity and let $\bar{h} = N^{-1} \mathbf{1}^\top \mathbf{M}(\boldsymbol{\rho}^{\max}) \mathbf{v}$, an upper bound on realized innovation. After innovation, the initial order is retained with probability $1 - \pi_\eta(z)$ and replaced by a uniformly random permutation with probability

$$\pi_\eta(z) = \frac{\eta z}{Y_0 + Dz}, \quad 0 \leq \eta \leq D + \frac{Y_0}{\bar{h}}. \quad (8)$$

Thus $\pi_\eta \in [0, 1]$ on the entire feasible domain and $\pi'_\eta(z) = \eta Y_0 / (Y_0 + Dz)^2 \geq 0$. Conditional on a redraw, the permutation is independent of the inputs. The draw changes consumption priority only; the installed graph and its productive capacities remain fixed.

The parameter η measures exposure to a redraw. Since $Dz/Y(z)$ is the share of resources that innovation created, (8) reads $\pi_\eta(z) = (\eta/D) Dz/Y(z)$: the probability of a redraw is proportional to that share. At $\eta = 0$ the order is permanent; at $\eta = D$ the probability of a redraw equals that share; larger values make the order more contestable. A redraw need not change the order. The uniform permutation reproduces the initial order with probability $1/N!$, so the order actually changes with probability $\pi_\eta(z)(1 - 1/N!)$. Payoffs depend only on the redraw probability and on expected shares after a redraw.

Fix the network ρ and the realized average innovation z , so the stock is $Y(z)$. With probability $1 - \pi_\eta(z)$ the initial order stands and group i consumes $a_i Y(z)$. With probability $\pi_\eta(z)$ the order is redrawn, group i lands on each rank with probability $1/N$, and it consumes $Y(z)/N$ on average. Its expected consumption given z is therefore

$$\begin{aligned} H_i(z) &= [1 - \pi_\eta(z)] a_i Y(z) + \pi_\eta(z) \frac{Y(z)}{N} = a_i Y(z) - \pi_\eta(z) Y(z) \left(a_i - \frac{1}{N} \right) \\ &= a_i Y_0 + \Delta_i z, \quad \Delta_i = D a_i - \eta \left(a_i - \frac{1}{N} \right), \end{aligned} \quad (9)$$

where the last step uses $Y(z) = Y_0 + Dz$ and $\pi_\eta(z) Y(z) = \eta z$ from (8). Before the inputs are drawn, z is random with mean $h(\rho)$ by (6). Since H_i is affine in z , expected consumption is $a_i Y_0 + \Delta_i h(\rho)$, and subtracting the link costs $K_i(\rho)$ from (2) gives the expected net utility

$$U_i(\rho) = a_i Y_0 + \Delta_i h(\rho) - K_i(\rho). \quad (10)$$

In particular, $\sum_i \Delta_i = D$. In Δ_i , the term $D a_i$ is the resources that a unit of innovation adds to group i 's claim, and $\eta(a_i - 1/N)$ is the expected status rent it puts at risk: η converts a group's advantage over the average share into an expected loss per unit of innovation.

3 Equilibrium

The equilibrium conditions compare the current network with changes made by one group or one pair.

Let $\mathcal{P} = \binom{V}{2}$ be the set of all possible pairs, including those that do not currently interact. We write $e = ij$ for the unordered pair $\{i, j\}$. Its intensity is $\rho_e = \rho_{ij}$, its allowed intensities form $\mathcal{R}_e = \mathcal{R}_{ij}$, and its capacity is $\bar{\rho}_e = \bar{\rho}_{ij}$. A network $\rho = (\rho_e)_{e \in \mathcal{P}}$ lists one intensity for every pair. The feasible networks form $\mathcal{R} = \prod_{e \in \mathcal{P}} \mathcal{R}_e$: this product means choosing one allowed intensity for each pair.

To test unilateral deviations, represent the current network by offers $\ell_{ij} = \ell_{ji} = \rho_{ij}$. If i alone changes its offer to $s \in \mathcal{R}_{ij}$, the new intensity is $\min\{s, \rho_{ij}\}$. A higher offer therefore leaves interaction unchanged; a lower offer reduces it. Increasing interaction to $t > \rho_{ij}$ requires both groups to offer at least t . Definition 1 tests unilateral reductions in condition (i) and mutually agreed increases in condition (ii).

The set $\mathcal{P}_i = \{e \in \mathcal{P} : i \in e\}$ contains the pairs involving group i . Group i cannot change links between other groups. Write $\tilde{\rho}$ for an alternative network. All networks that i can obtain through unilateral offer changes form

$$\mathcal{D}_i(\rho) = \{\tilde{\rho} \in \mathcal{R} : \tilde{\rho}_e \leq \rho_e \ \forall e \in \mathcal{P}_i, \quad \tilde{\rho}_e = \rho_e \ \forall e \in \mathcal{P} \setminus \mathcal{P}_i\}. \quad (11)$$

The first restriction lets i reduce any number of its own links; the second leaves all other links unchanged. Requiring $\tilde{\rho} \in \mathcal{R}$ keeps every intensity feasible.

For a pair e and an allowed intensity $t \in \mathcal{R}_e$, let $\boldsymbol{\rho}^{e,t}$ denote the network that sets the intensity of e to t and leaves every other pair unchanged. For either group $r \in \{i, j\}$ in pair $e = ij$, define

$$\Gamma_r(e, t; \boldsymbol{\rho}) = U_r(\boldsymbol{\rho}^{e,t}) - U_r(\boldsymbol{\rho}),$$

where U_r is expected net utility from (10). Thus Γ_r is the change in payoff: positive means a gain, negative means a loss, and zero means indifference. The bilateral condition below considers increases, $t > \rho_e$, which require both groups' agreement.

Definition 1 (Pairwise Nash stability). A network $\boldsymbol{\rho} \in \mathcal{R}$ is an equilibrium if and only if both conditions hold:

(i) For every $i \in V$,

$$U_i(\boldsymbol{\rho}) \geq U_i(\tilde{\boldsymbol{\rho}}) \quad \forall \tilde{\boldsymbol{\rho}} \in \mathcal{D}_i(\boldsymbol{\rho}). \quad (12)$$

(ii) For every $e = ij \in \mathcal{P}$ and every $t \in \mathcal{R}_e$ with $t > \rho_e$,

$$\min\{\Gamma_i(e, t; \boldsymbol{\rho}), \Gamma_j(e, t; \boldsymbol{\rho})\} < 0 \quad \text{or} \quad \Gamma_i(e, t; \boldsymbol{\rho}) = \Gamma_j(e, t; \boldsymbol{\rho}) = 0. \quad (13)$$

Condition (12) rules out profitable simultaneous reductions or deletions of any of a group's links. Under the canonical offers $\ell_{ij} = \rho_{ij}$, unilateral offer deviations generate exactly $\mathcal{D}_i(\boldsymbol{\rho})$, so these offers are a Nash equilibrium if and only if (12) holds. Condition (13) rules out a link increase that makes both groups weakly better off and at least one strictly better off. It includes creating a link from zero and allows both groups to be indifferent. No transfers or joint changes to other links are permitted in this bilateral test.

The offer game always has a Nash equilibrium at zero offers, because no group can create a link alone. The existence result below also requires stability against bilateral increases.

For any set of links $E \subseteq \mathcal{P}$, the *saturated network* $\boldsymbol{\rho}^E$ puts every link in E at its capacity and every other pair at zero. Write $h(E)$ for its expected innovation and E_i for the links in E that involve group i :

$$(\boldsymbol{\rho}^E)_e = \bar{\rho}_e \mathbf{1}\{e \in E\}, \quad h(E) := h(\boldsymbol{\rho}^E), \quad E_i := E \cap \mathcal{P}_i.$$

Lemma 1 (Productive complementarity). *Extend h by (6) to the box $\prod_{e \in \mathcal{P}} [0, \bar{\rho}_e]$. At every point of this box,*

$$\begin{aligned} \frac{\partial h}{\partial \rho_e} &> 0, \quad \frac{\partial^2 h}{\partial \rho_e^2} > 0 \quad \forall e \in \mathcal{P}, \\ \frac{\partial^2 h}{\partial \rho_e \partial \rho_f} &\geq 0 \quad \forall e, f \in \mathcal{P}, \quad e \neq f. \end{aligned}$$

Thus h is strictly increasing and strictly convex in each edge intensity and has increasing differences across intensities. In particular, for every $A \subseteq B \subseteq \mathcal{P}$ and $e \in \mathcal{P} \setminus B$,

$$h(A \cup \{e\}) - h(A) \leq h(B \cup \{e\}) - h(B). \quad (14)$$

Strengthening a link raises innovation at an increasing rate, holding other links fixed. Other links can increase this gain. Thus, adding the same link yields at least as much additional innovation when the network already contains more links, as in inequality (14).

Proof. The capacities $\bar{\rho}_e$, and hence $\bar{\rho} = \max_e \bar{\rho}_e$, are fixed, as are β and $\mathbf{v}$. Write $\mathbf{G} = \mathbf{G}(\boldsymbol{\rho})$ and $\mathbf{M} = \mathbf{M}(\boldsymbol{\rho})$. The bound $\beta(N-1) < 1$ holds throughout the box, so $\mathbf{I} - \beta\mathbf{G}$ is invertible there; its inverse is smooth on an open neighborhood of the box, so the derivatives below also apply at zero and at capacity.

For $e = ij$, let $\mathbf{S}_e$ be the constant 0–1 matrix with ones exactly in entries (i, j) and (j, i) . Then $\mathbf{G} = \bar{\rho}^{-1} \sum_{f \in \mathcal{P}} \rho_f \mathbf{S}_f$, so $\partial \mathbf{G} / \partial \rho_e = \mathbf{S}_e / \bar{\rho}$. Differentiating $(\mathbf{I} - \beta\mathbf{G})\mathbf{M} = \mathbf{I}$ and multiplying on the left by $\mathbf{M}$ gives

$$\frac{\partial \mathbf{M}}{\partial \rho_e} = \frac{\beta}{\bar{\rho}} \mathbf{M} \mathbf{S}_e \mathbf{M}.$$

Since only $\mathbf{M}$ depends on ρ_e in $h = N^{-1} \mathbf{1}^\top \mathbf{M} \mathbf{v}$, applying this formula once and then again to each factor $\mathbf{M}$ yields, for all $e, f \in \mathcal{P}$ (including $f = e$),

$$\frac{\partial h}{\partial \rho_e} = \frac{\beta}{N\bar{\rho}} \mathbf{1}^\top \mathbf{M} \mathbf{S}_e \mathbf{M} \mathbf{v}, \quad \frac{\partial^2 h}{\partial \rho_f \partial \rho_e} = \frac{\beta^2}{N\bar{\rho}^2} \mathbf{1}^\top (\mathbf{M} \mathbf{S}_f \mathbf{M} \mathbf{S}_e \mathbf{M} + \mathbf{M} \mathbf{S}_e \mathbf{M} \mathbf{S}_f \mathbf{M}) \mathbf{v}.$$

Every factor has nonnegative entries, so the cross partials are nonnegative. For strict positivity, the series in (5) gives $\mathbf{M} \geq \mathbf{I}$ entry by entry. Replacing each $\mathbf{M}$ by $\mathbf{I}$ therefore gives

$$\mathbf{1}^\top \mathbf{M} \mathbf{S}_e \mathbf{M} \mathbf{v} \geq \mathbf{1}^\top \mathbf{S}_e \mathbf{v} = v_i + v_j > 0, \quad \mathbf{1}^\top \mathbf{M} \mathbf{S}_e \mathbf{M} \mathbf{S}_e \mathbf{M} \mathbf{v} \geq \mathbf{1}^\top \mathbf{S}_e^2 \mathbf{v} = v_i + v_j > 0,$$

because $\mathbf{S}_e$ swaps coordinates i and j and $\mathbf{S}_e^2$ restores them. Hence h is strictly increasing and strictly convex in each intensity, and has increasing differences across intensities.

We now prove (14). For $C \in \{A, B\}$ and $t \in [0, \bar{\rho}_e]$, let $\boldsymbol{\rho}_C(t)$ set e at t , the links in C at capacity, and all others at zero, and let $g_C(t) = h(\boldsymbol{\rho}_C(t))$. Then $g_C(0) = h(C)$, $g_C(\bar{\rho}_e) = h(C \cup \{e\})$, and $g'_C(t) = \partial h / \partial \rho_e(\boldsymbol{\rho}_C(t))$. Fix t and pass from $\boldsymbol{\rho}_A(t)$ to $\boldsymbol{\rho}_B(t)$ by raising the links in $B \setminus A$ one at a time from zero to capacity. If $\mathbf{x}(u)$ denotes the network when the current link f has intensity u ,

$$\frac{\partial h}{\partial \rho_e}(\mathbf{x}(\bar{\rho}_f)) - \frac{\partial h}{\partial \rho_e}(\mathbf{x}(0)) = \int_0^{\bar{\rho}_f} \frac{\partial^2 h}{\partial \rho_f \partial \rho_e}(\mathbf{x}(u)) du \geq 0,$$

so $g'_B(t) \geq g'_A(t)$ for every t . Integrating over $[0, \bar{\rho}_e]$,

$$h(B \cup \{e\}) - h(B) = \int_0^{\bar{\rho}_e} g'_B(t) dt \geq \int_0^{\bar{\rho}_e} g'_A(t) dt = h(A \cup \{e\}) - h(A).$$

We also need a property used later: the innovation loss from deleting links weakly increases with β and with each v_i . Fix $F \subseteq E \subseteq \mathcal{P}$ and put $\mathbf{G}_E = \mathbf{G}(\boldsymbol{\rho}^E)$ and $\mathbf{G}_{E \setminus F} = \mathbf{G}(\boldsymbol{\rho}^{E \setminus F})$. The $k = 0$ terms of (5) cancel, so

$$h(E) - h(E \setminus F) = \frac{1}{N} \sum_{k \geq 1} \beta^k \mathbf{1}^\top (\mathbf{G}_E^k - \mathbf{G}_{E \setminus F}^k) \mathbf{v},$$

and, by telescoping,

$$\mathbf{G}_E^k - \mathbf{G}_{E \setminus F}^k = \sum_{m=0}^{k-1} \mathbf{G}_E^{k-1-m} (\mathbf{G}_E - \mathbf{G}_{E \setminus F}) \mathbf{G}_{E \setminus F}^m \geq \mathbf{0}.$$

Every factor is nonnegative because deleting links only lowers entries of $\mathbf{G}$, and none depends on β or $\mathbf{v}$. Thus raising β or any v_i weakly raises each term in the loss series. $\square$

Proposition 1 (Isolation and saturation). *Recall from (9) that $\Delta_i = Da_i - \eta(a_i - 1/N)$ is the change in group i 's expected consumption per unit of innovation, before interaction costs. Every equilibrium $\rho \in \mathcal{R}$ satisfies*

$$\Delta_i \leq 0 \implies \rho_e = 0 \quad \forall e \in \mathcal{P}_i, \quad \forall i \in V,$$

and

$$\rho_e \in \{0, \bar{\rho}_e\} \quad \forall e \in \mathcal{P}.$$

Hence every group with $\Delta_i \leq 0$ is isolated, and every active link is at capacity.

Proof. If $\Delta_i \leq 0$, deleting a positive incident link weakly raises gross utility and saves a strictly positive cost. Thus i must be isolated. For an active edge at intensity $t > 0$, fix all other links and write $\phi(t)$ for expected innovation as a function of that intensity alone. Equilibrium requires each endpoint r to prefer retaining the edge to deleting it, $\Delta_r[\phi(t) - \phi(0)] \geq \kappa_0 + \kappa_1 t$, which forces $\Delta_r > 0$. Strict convexity of ϕ (Lemma 1) then gives

$$\Delta_r \phi'(t) > \Delta_r \frac{\phi(t) - \phi(0)}{t} \geq \kappa_1 + \frac{\kappa_0}{t} > \kappa_1.$$

If $t < \bar{\rho}_e$, a sufficiently small feasible increase strictly benefits both endpoints, a contradiction. This includes the lower endpoint of any nondegenerate positive band. A singleton band is already saturated. $\square$

The conclusion uses affine innovation values and the specified cost function; it need not survive concave private benefits or convex marginal interaction costs. The veto in the first display is not selective. A group with $\Delta_i \leq 0$ rejects every partner, whatever the partner's rank, because its exposure depends on aggregate innovation and not on who produces it.

For $e = ij \in \mathcal{P}$, let $c_e = \kappa_0 + \kappa_1 \bar{\rho}_e$ be the cost of the link at capacity to each endpoint. For $e \notin E$, adding e at capacity to ρ^E gives $\rho^{E \cup \{e\}}$, so the payoff change Γ_r of each endpoint $r \in e$ reduces to

$$\Gamma_r(e; E) := \Gamma_r(e, \bar{\rho}_e; \rho^E) = \Delta_r[h(E \cup \{e\}) - h(E)] - c_e. \quad (15)$$

Theorem 1 (Exact characterization and existence). *A saturated network ρ^E , with $E \subseteq \mathcal{P}$, is an equilibrium if and only if both tests hold:*

$$\Delta_i[h(E) - h(E \setminus F)] \geq \sum_{e \in F} c_e \quad \forall i \in V, \quad \forall F \subseteq E_i, \quad (16)$$

$$\min\{\Gamma_i(e; E), \Gamma_j(e; E)\} < 0 \quad \text{or} \quad \Gamma_i(e; E) = \Gamma_j(e; E) = 0 \quad \forall e = ij \in \mathcal{P} \setminus E. \quad (17)$$

An equilibrium exists for every admissible parameter vector. Moreover, there is a greatest equilibrium $E^{\max}$ under edge inclusion, and every equilibrium is a subset of $E^{\max}$.

This is a finite solution to the continuous game. There are $N(N-1)/2$ pairs and one saturated network for each subset of them, $2^{N(N-1)/2}$ in all, and each requires one linear-system solution for $h(E)$. The test includes the empty network and equality cases. In (17), a pair with one indifferent endpoint and one strictly benefiting endpoint blocks the current network by adding the missing link.

Call a network *deletion-stable* if it satisfies (16), and let

$$\mathfrak{D} = \{E \subseteq \mathcal{P} : E \text{ satisfies (16)}\}.$$

The existence proof establishes

$$\emptyset \in \mathfrak{D}, \quad A, B \in \mathfrak{D} \implies A \cup B \in \mathfrak{D}, \quad E^{\max} = \bigcup_{E \in \mathfrak{D}} E,$$

and shows that $\rho^{E^{\max}}$ is an equilibrium. Existence follows from these finite unions and the complementarity in Lemma 1.

Proof. We first prove the characterization. Necessity of (16)–(17) follows by considering complete deletion subsets and saturated additions. For sufficiency, a network satisfying (16) cannot give an incident link to $\Delta_i \leq 0$. Fix a group with incident edges and $\Delta_i > 0$. Holding its other intensities fixed, compare a feasible partial level $0 < t < u = \bar{\rho}_e$ with endpoints $0, u$. Its one-coordinate payoff V satisfies

$$V(t) < (1 - t/u)V(0) + (t/u)V(u).$$

Indeed, $\Delta_i h$ is strictly convex, the variable cost is linear, and at $t > 0$ the group pays the whole fixed cost, rather than the fraction $(t/u)\kappa_0$ entering the convex combination. Constants and costs of other links cancel. Starting from any unilateral deviation, replace its partial coordinates one at a time by whichever of $0, u$ has the higher payoff. This never lowers payoff and ends at a deletion subset of the original saturated network. Condition (16) therefore rules out every profitable unilateral deviation.

For a missing edge, any partial addition that weakly benefits both endpoints requires both coefficients to be positive, since its cost is positive and its innovation gain is positive. Applying the strict convex-combination inequality to each endpoint shows that the saturated addition strictly benefits both. Condition (17) rules this out. Present edges cannot be strengthened. These observations exhaust the allowed deviations.

We now prove existence and identify the greatest equilibrium. The empty graph is deletion-stable. Any deletion-stable graph isolates all groups with $\Delta_i \leq 0$, as its one-edge deletion condition would otherwise fail. Let A and B be deletion-stable and write $U = A \cup B$. For an incident deletion set $F \subseteq U_i$, partition it into $F_1 = F \setminus B \subseteq A_i$ and $F_2 = F \cap B \subseteq B_i$. Since $U \supseteq A$ and $U \setminus F_1 \supseteq B$, increasing differences implies

$$\begin{aligned} h(U) - h(U \setminus F) &= [h(U) - h(U \setminus F_1)] + [h(U \setminus F_1) - h(U \setminus F)] \\ &\geq [h(A) - h(A \setminus F_1)] + [h(B) - h(B \setminus F_2)]. \end{aligned}$$

For $\Delta_i > 0$, multiplying by Δ_i and applying the two deletion inequalities bounds this below by $\sum_{e \in F} c_e$. Groups with nonpositive coefficients have no incident edges. Thus U is deletion-stable. Finite repeated unions give a greatest deletion-stable graph $E^{\max}$.

Suppose a missing edge $e = ij$ weakly benefits both endpoints when added to $E^{\max}$. Its positive cost forces $\Delta_i, \Delta_j > 0$. For any group, a deviation deleting old edges but retaining e becomes no more attractive after its addition, by increasing differences. For an endpoint's deletion that also removes e , its payoff after deletion is $U_i(E^{\max} \setminus F)$ for some old incident set F . This is at most $U_i(E^{\max})$, which is at most $U_i(E^{\max} \cup \{e\})$. Hence the larger graph is deletion-stable, a contradiction. In particular, no missing edge blocks $E^{\max}$, which is an equilibrium by the finite characterization. Every equilibrium is deletion-stable and is therefore contained in $E^{\max}$. $\square$

Several implications distinguish the model from one based only on reachability. Equilibrium can contain cycles because another weighted walk can raise innovation even when its endpoints are already connected. Positive lower bounds do not affect equilibrium networks when upper bounds are held fixed. Finally, checking each edge separately is insufficient: joint deletion can eliminate complementary links whose individual retention tests pass.

4 Comparative statics

4.1 Productive returns, costs, and appropriation rents

Proposition 2 (Monotone changes in the greatest equilibrium). *Hold the capacity bands and their normalization fixed. On any common admissible parameter domain, $E^{\max}$ weakly expands by inclusion under each of the following changes, holding other parameters fixed:*

- (i) κ_0 or κ_1 falls, with κ_0 remaining positive.
- (ii) β rises.
- (iii) $\mathbf{v}$ rises componentwise.
- (iv) D rises, holding shares and η fixed.

More generally, raising every Δ_i while holding technology and costs fixed weakly expands $E^{\max}$. Hence integration in the greatest equilibrium weakly rises in each comparison.

Proof. For a deletion-stable graph, all groups with incident links have positive coefficients. Each change preserves the deletion inequalities in (16):

- (i) Lower κ_0 or κ_1 reduces the costs saved by deleting links, while leaving the innovation loss unchanged.
- (ii) By Lemma 1, a larger β weakly increases the innovation loss from any fixed deletion, while leaving costs unchanged.
- (iii) The same lemma shows that raising $\mathbf{v}$ componentwise weakly increases the innovation loss from any fixed deletion, while leaving costs unchanged.
- (iv) A rise in D increases every Δ_i by a_i times the change in D , while leaving technology and costs unchanged. More generally, raising every Δ_i preserves the deletion inequalities.

Thus the old greatest equilibrium is deletion-stable after each change and is contained in the new greatest one. Fixed capacities turn inclusion into weakly greater integration. $\square$

The result concerns the greatest equilibrium; it does not impose a selection rule on other equilibria. A larger baseline endowment Y_0 has no effect on incentives under (8), although it enlarges the admissible range of η .

Exposure changes incentives in opposite directions: $\partial\Delta_i/\partial\eta = 1/N - a_i$. For an advantaged group, solving $\Delta_i \leq 0$ gives the sufficient threshold for isolation

$$\eta \geq \eta_i^{\text{veto}} = \frac{Da_i}{a_i - 1/N}, \quad a_i > 1/N. \quad (18)$$

Proposition 1 applies whenever this exposure is admissible, which requires $Y_0/\bar{h} \geq D/(Na_i - 1)$. The threshold falls with the group's initial share; costs can induce withdrawal earlier. At $\eta = D$, all groups have $\Delta_i = D/N$. Above that level, the groups with larger shares value innovation less.

4.2 Withdrawal and the loss of complementary links

To isolate network transmission, hold all primitives fixed and vary only the set of pairs allowed to interact. For a set of available pairs $Q \subseteq \mathcal{P}$, permit positive interaction only on Q and denote the greatest equilibrium by $E^{\max}(Q)$. Withdrawal of groups $W \subseteq V$ means $Q = \binom{V \setminus W}{2}$: the groups remain in the economy with their own inputs and consumption shares, but cannot interact. In particular, neither N nor $\bar{\rho}$ is recomputed.

Proposition 3 (Induced withdrawal). *Let $E = E^{\max}(\mathcal{P})$ and $R = E \cap Q$, the network after unavailable links are removed. Then:*

- (i) *The restricted game has a greatest equilibrium, and $E^{\max}(Q) \subseteq R$.*
- (ii) *Further links are lost, $E^{\max}(Q) \subsetneq R$, if and only if some group i and nonempty $F \subseteq R_i$ satisfy*

$$\Delta_i[h(R) - h(R \setminus F)] < \sum_{e \in F} c_e. \quad (19)$$

Proof. (i) The union and addition arguments in Theorem 1 apply to Q . Every restricted deletion-stable graph is also deletion-stable in the original game, since deletion tests do not depend on absent links. It is therefore contained in both E and Q , hence in R .

- (ii) If R passes every deletion test, it is contained in the greatest restricted equilibrium, so (i) gives equality. If a test fails, R cannot be an equilibrium, and the inclusion is strict. □

In E , the benefit of retaining F covered its cost. Removing other links lowers that benefit through complementarity. Further withdrawal occurs exactly when it falls below cost, as in (19). With innovation additive across links, the benefit would be unchanged and no further withdrawal would occur. This compares equilibria while holding exposure fixed; it does not specify an adjustment process.

Example 2 (Withdrawal in a three-group network). Consider three groups with capacities $(\bar{\rho}_{12}, \bar{\rho}_{13}, \bar{\rho}_{23}) = (1, 1, 1/2)$, $\beta = 1/4$, deterministic inputs $\xi = \mathbf{1}$, and $(\kappa_0, \kappa_1) = (1/20, 0)$. Let $D = 1$, $Y_0 = 4$, and shares be $(1/2, 1/3, 1/6)$. At $\eta = 1$, every $\Delta_i = 1/3$; all deletion tests at the complete graph hold strictly, so it is the greatest equilibrium. If group 1 withdraws, $R = \{23\}$ and

$$h(R) = \frac{23}{21}, \quad h(\emptyset) = 1, \quad \Delta_2[h(R) - h(\emptyset)] = \frac{2}{63} < \frac{1}{20}. \quad (20)$$

The restricted equilibrium is uniquely empty: two groups with positive innovation values also become isolated. This can arise endogenously from greater exposure. Since $\bar{h} = 31/18$, the admissible bound is $103/31 > 3$. At $\eta = 3$, the coefficients are $(0, 1/3, 2/3)$. Group 1 must be isolated, and group 2 still rejects pair 23, so every group is isolated even though the two remaining groups' innovation values have weakly increased.

Greater exposure can both remove partners and change the remaining groups' own incentives. Let $\eta < \eta'$ be admissible, let E be the greatest equilibrium at η , and write Δ_i, Δ'_i for the two innovation values. Let Z be any set of groups isolated in every equilibrium at η' , such as those with $\Delta'_i \leq 0$, and define

$$R = E \setminus \bigcup_{i \in Z} E_i. \quad (21)$$

For $i \notin Z$ and nonempty $F \subseteq R_i$, write

$$\begin{aligned} m_i(F) &= \Delta_i[h(E) - h(E \setminus F)] - \sum_{e \in F} c_e, \\ \kappa_i(F) &= \Delta_i\{[h(E) - h(E \setminus F)] - [h(R) - h(R \setminus F)]\}, \\ \iota_i(F) &= (\eta' - \eta)(a_i - 1/N)[h(R) - h(R \setminus F)]. \end{aligned} \tag{22}$$

These terms are the original retention margin, the return lost with partners' links, and the loss due to the group's changed valuation, respectively. The last term is negative when exposure raises the group's valuation.

Proposition 4 (Withdrawal induced by greater exposure). *For every $i \notin Z$ and nonempty $F \subseteq R_i$,*

$$\Delta'_i[h(R) - h(R \setminus F)] - \sum_{e \in F} c_e = m_i(F) - \kappa_i(F) - \iota_i(F), \tag{23}$$

with $m_i(F), \kappa_i(F) \geq 0$. Moreover:

- (a) *If $m_i(F) \geq \kappa_i(F) + \iota_i(F)$ for every such i, F , then R is contained in the greatest equilibrium at η' .*
- (b) *If a test fails, R is not an equilibrium at η' . Every equilibrium at η' contained in E is then a strict subset of R . This applies to the greatest equilibrium when E contains every pair outside Z .*
- (c) *If a test fails for a group with $a_i \leq 1/N$, then $\kappa_i(F) > m_i(F)$: lost complementary returns alone would induce withdrawal at the old innovation values.*

Proof. *Identity.* Substitute $\Delta'_i = \Delta_i - (\eta' - \eta)(a_i - 1/N)$ and add and subtract $\Delta_i[h(E) - h(E \setminus F)]$. The old deletion test gives $m_i(F) \geq 0$. Since i has a link in E , $\Delta_i > 0$; increasing differences gives

$$h(E) - h(E \setminus F) \geq h(R) - h(R \setminus F),$$

so $\kappa_i(F) \geq 0$.

- (a) The identity gives every deletion inequality for R at η' . By Theorem 1, R is contained in the greatest equilibrium.
- (b) A failed test violates deletion stability. Every new equilibrium contained in E has no links at Z , so it is contained in R and cannot equal it. If E contains every pair outside Z , every new equilibrium is contained in E .
- (c) Here $\iota_i(F) \leq 0$. A failed test therefore implies $\kappa_i(F) > m_i(F)$, equivalently (19) at the old values with $Q = \binom{V \setminus Z}{2}$.

□

The qualification in (b) matters: greater exposure can also create new links among disadvantaged groups. In Example 2, group 2 holds the average share, so its valuation does not change. Its withdrawal is entirely due to the loss of group 1's links.

Ordered capacities. For $1 \leq k \leq N-1$, if $\Delta_k \leq 0$, then $\eta > D$ and $\Delta_i \leq \Delta_k$ for $i \leq k$. Thus the first k groups are isolated, removing at least the capacity share

$$S(\boldsymbol{\rho}) \geq \frac{\sum_{r=1}^k (N-r)\bar{\rho}_r}{\sum_{r=1}^{N-1} (N-r)\bar{\rho}_r}. \quad (24)$$

There are $N-r$ pairs whose more powerful endpoint is r . When $k \leq N-2$, each removed pair has greater capacity than every remaining pair, so this share exceeds the fraction $k(2N-k-1)/[N(N-1)]$ of pairs removed. Ordered capacities therefore concentrate the loss in the strongest links.

4.3 Full integration with any number of groups

Hold all primitives except η fixed and write $\bar{\eta} = D + Y_0/\bar{h}$ for its admissible upper bound. For each group define

$$\tau_i = \max_{\emptyset \neq F \subseteq \mathcal{P}_i} \frac{\sum_{e \in F} c_e}{h(\mathcal{P}) - h(\mathcal{P} \setminus F)}. \quad (25)$$

Each denominator is strictly positive by Lemma 1. The threshold τ_i is the smallest private innovation value that deters every deletion by group i from the complete network. It accounts for joint deletions, not just individual links.

Corollary 1 (A full-integration interval for any number of groups). *For any $N \geq 2$, define*

$$\begin{aligned} \eta_L &= \max \left\{ 0, \max_{i: a_i < 1/N} \frac{\tau_i - Da_i}{1/N - a_i} \right\}, \\ \eta_U &= \min \left\{ \bar{\eta}, \min_{i: a_i > 1/N} \frac{Da_i - \tau_i}{a_i - 1/N} \right\}. \end{aligned} \quad (26)$$

If $\tau_i \leq D/N$ for every group with $a_i = 1/N$ and $\eta_L \leq \eta_U$, the complete network at capacity is an equilibrium exactly for $\eta \in [\eta_L, \eta_U]$. Otherwise it is never an equilibrium. The interval may be a single point. Equivalently, the greatest equilibrium has $C = 1$ exactly on this interval; outside it, every equilibrium has $C < 1$.

Proof. (i) *Retention.* The complete network has no missing links. Its deletion tests are equivalent to $\Delta_i(\eta) \geq \tau_i$ for every i , by dividing by each positive innovation loss and maximizing over deletion sets.

(ii) *Exposure bounds.* Since $\Delta_i(\eta) = Da_i + \eta(1/N - a_i)$, disadvantaged groups impose lower bounds on η , advantaged groups impose upper bounds, and average-share groups require $\tau_i \leq D/N$. Intersecting these conditions with $[0, \bar{\eta}]$ gives (26). Full integration has $C = 1$; every other network has $C < 1$. □

Full integration can appear and disappear at most once as exposure rises. This does not order partially integrated equilibria.

Example 3 (Two groups). Let shares be $(b, 1-b)$, $1/2 < b < 1$, and mean inputs be $\mathbf{v} = v\mathbf{1}$, $v > 0$. Write u for link capacity, $g = u/\bar{\rho} = 1$, and $c = \kappa_0 + \kappa_1 u$. The innovation gain from the link and the smaller private value are

$$L = \frac{v\beta g}{1 - \beta g}, \quad m(\eta) = \min\{\Delta_1, \Delta_2\} = \frac{D}{2} - (b - \tfrac{1}{2})|\eta - D|. \quad (27)$$

Both groups retain the link exactly when $m(\eta)L \geq c$. If $c/L < D/2$, this gives

$$|\eta - D| \leq \frac{D/2 - c/L}{b - 1/2}, \quad (28)$$

intersected with the admissible exposure range. The weaker group limits cooperation at low exposure and the stronger group at high exposure. The linked network is the unique equilibrium in this interval: an absent link benefits both groups, or benefits one while leaving the other indifferent. Outside it, only the empty network is an equilibrium. If $c/L = D/2$, both networks are equilibria only at $\eta = D$; at other exposures, or if $c/L > D/2$, only the empty network remains.

5 Welfare and a four-group economy

5.1 Efficiency and compensation

Summing utilities and using $\sum_i \Delta_i = D$ gives welfare:

$$W(\rho) = Y_0 + Dh(\rho) - 2 \sum_{i < j} (\kappa_0 \mathbf{1}\{\rho_{ij} > 0\} + \kappa_1 \rho_{ij}). \quad (29)$$

The lottery changes consumption shares but not their sum, so the planner's choice is independent of η . With a partial link $0 < t < \bar{\rho}_e$, coordinatewise convexity and the fixed cost imply that either deleting it or raising it to capacity strictly improves welfare. The planner therefore solves the finite problem

$$\arg \max_{E \subseteq \mathcal{P}} \left\{ Y_0 + Dh(E) - 2 \sum_{e \in E} c_e \right\}. \quad (30)$$

For two groups, the link is strictly efficient when $DL > 2c$, even at exposures where one group refuses it. No general ordering of equilibrium and efficient edge sets is established for larger networks. The example below calculates the welfare loss directly.

Cost reimbursement. If $\Delta_i < 0$, deleting a link raises group i 's expected consumption. Reimbursing its entire interaction cost cannot remove that incentive. This claim requires a strict consumption loss: when $\Delta_i = 0$, full reimbursement can leave the group indifferent.

Compensating transfers. An enforceable allocation $t_i + Dz/N$, with $\sum_i t_i = Y_0$, would give each group the positive innovation value D/N at every exposure. In Example 4, this sustains the efficient complete network. At $\eta = 6$, suitable intercepts t_i can also leave every group better off than in the original equilibrium: the intercepts that leave all groups indifferent sum to $Y_0 - 2.34$, leaving a surplus of 2.34 to distribute.

Such transfers require a commitment that survives changes in power. The political-losers problem concerns lost claims on resources (Acemoglu, 2025); paying interaction costs does not compensate for those losses. Following the commitment problem in Acemoglu (2003), the benchmark assumes that no contract can override allocation by realized priority. Institutions that constrain future rulers can support such commitments (North and Weingast, 1989). Allowing them changes this assumption and, in the four-group example, recovers the welfare loss.

5.2 A four-group economy

Example 4 (A four-group economy). Consider four groups with deterministic inputs $\xi = \mathbf{1}$ and

$$Y_0 = 10, \quad D = 2, \quad (a_1, a_2, a_3, a_4) = (0.45, 0.35, 0.15, 0.05), \quad \beta = \frac{1}{4}, \quad \kappa_0 = \frac{1}{40}, \quad \kappa_1 = 0. \quad (31)$$

The three power-class bands are $[0.8, 1]$, $[0.4, 0.6]$, and $[0.1, 0.3]$. Pairs involving group 1 have capacity 1, pairs 23 and 24 have capacity 0.6, and pair 34 has capacity 0.3. Total capacity is 4.5 and $\bar{h} = 2.337281$, so all exposures in Table 1 are below the admissible bound 6.278476.

Table 1: Unique equilibrium at three levels of status exposure

η	$(\Delta_1, \Delta_2, \Delta_3, \Delta_4)$	Active pairs	C	h	W
0	(0.9, 0.7, 0.3, 0.1)	12, 13, 14, 23, 24	0.9333	2.1764	14.1027
2	(0.5, 0.5, 0.5, 0.5)	All six	1	2.3373	14.3746
6	(-0.3, 0.1, 0.9, 1.3)	34	0.0667	1.0405	12.0311

Every active link is at capacity. All 64 saturated networks are checked using the exact conditions in Theorem 1, including every incident-edge deletion subset. Computation uses rational arithmetic; reported values are rounded. The unique planner solution is the full network at all three values of η .

At $\eta = 0$, group 4 refuses its weakest link, 34. At $\eta = 2$, all groups have the same innovation value and the network is complete. At $\eta = 6$, group 1 vetoes all interaction and group 2 also withdraws, leaving only 34. Innovation is then 55.5% below the efficient level and welfare is 16.3% lower.

Group 2's exit involves both channels in Proposition 4. Between $\eta = 2$ and $\eta = 6$, take $Z = \{1\}$, $R = \{23, 24, 34\}$, and $F = \{23, 24\}$. The original margin is $m_2 \approx 0.224$, the lost complementary return is $\kappa_2 \approx 0.168$, and the incentive loss is $\iota_2 \approx 0.084$. Their combined effect gives $m_2 - \kappa_2 - \iota_2 \approx -0.029$. It would retain those links at its old value in R (surplus 0.055), or at its new value in the complete network (surplus 0.005). Together, the loss of its partner and the fall in its own value make it withdraw. Values are rounded.

Corollary 1 gives the full-integration interval

$$\eta \in \left[\frac{109831}{396750}, \frac{4469699}{1075830} \right] \approx [0.276827, 4.154652]. \quad (32)$$

The lower endpoint binds group 4's retention of 34; the upper endpoint binds group 1's joint retention of all its links. Figure 1 shows the greatest equilibrium as exposure rises. These are model calculations, not calibrated magnitudes.

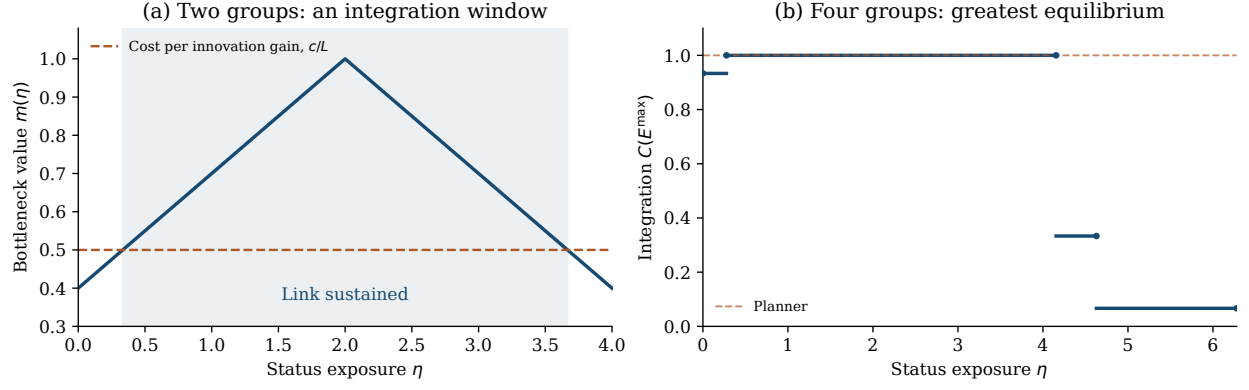


Figure 1: Status exposure and interaction. Panel (a) uses two groups with deterministic inputs, $D = 2$, $b = 0.8$, $v = 1$, $\beta = 1/4$, $g = 1$, and $c = 1/6$: the link is sustained for $\eta \in [1/3, 11/3]$. Then $\bar{h} = 4/3$, and every exposure in the plotted range is admissible when $Y_0 \geq 8/3$. Panel (b) uses the four-group economy in (31). Each horizontal segment gives its greatest equilibrium on an interval of η ; dots give values at interval boundaries. All change points are computed with exact rational arithmetic.

6 Robustness to the status lottery

The benchmark lottery makes two assumptions. The probability of a redraw is (8), and a redraw assigns ranks by a uniform permutation, which gives every group the expected share $1/N$. This section relaxes each in turn.

6.1 Who gains from a redraw

Keep (8), but let the new order be drawn from any distribution over the $N!$ orders that is independent of the inputs. Let p_{ir} be the probability that group i lands on rank r and

$$\hat{a}_i = \sum_{r=1}^N p_{ir} a_r, \quad \sum_{i=1}^N \hat{a}_i = 1, \quad (33)$$

its expected share after a redraw. The shares sum to one because every rank is filled. Call $a_i - \hat{a}_i$ the group's status rent. The steps that lead to (9) now give

$$H_i(z) = a_i Y_0 + \hat{\Delta}_i z, \quad \hat{\Delta}_i = D a_i - \eta(a_i - \hat{a}_i), \quad \sum_{i=1}^N \hat{\Delta}_i = D. \quad (34)$$

Expected consumption remains affine. Substituting $\hat{\Delta}_i$ for Δ_i therefore preserves saturation, the equilibrium characterization, and the comparative statics for costs, productivity, and withdrawal. Welfare is unchanged because the coefficients still sum to D .

The full-integration condition becomes $\hat{\Delta}_i(\eta) \geq \tau_i$ for every group. Each is a linear inequality in η , so their intersection with the admissible domain is again a possibly empty closed interval. A group with $a_i > \hat{a}_i$ vetoes interaction once the admissible exposure reaches $D a_i / (a_i - \hat{a}_i)$. Thus the mechanism requires a threatened resource share, not equalization after a redraw. The order of veto thresholds can change because it depends on the proportional loss $(a_i - \hat{a}_i)/a_i$.

6.2 The probability of a redraw

Retain the same technology, input distribution, consumption shares, and uniform redraw of priority, but replace (8) by an increasing C^1 function $\pi : [0, \bar{h}] \rightarrow [0, 1]$ with $\pi(0) = 0$. Expected consumption conditional on innovation is now

$$H_i(z) = (Y_0 + Dz) [a_i - \pi(z)(a_i - 1/N)].$$

Expected utility is $U_i(\boldsymbol{\rho}) = \mathbb{E}[H_i(z)] - K_i(\boldsymbol{\rho})$, where the distribution of z depends on $\boldsymbol{\rho}$ through (4); the equilibrium definition is unchanged. Define the marginal stock exposed to a redraw by $A_\pi(z) = D\pi(z) + (Y_0 + Dz)\pi'(z) = [(Y_0 + Dz)\pi(z)]'$. Then

$$H'_i(z) = Da_i - (a_i - 1/N)A_\pi(z). \quad (35)$$

The first term is the productive gain attached to the group's share. The second is the marginal loss of status rents. For $a_i \leq 1/N$, an increasing lottery gives $H'_i(z) > 0$: the prospect of redistribution reinforces the productive gain. For an advantaged group it can overturn that gain.

Proposition 5 (Robust isolation). *For a group with $a_i > 1/N$:*

(i) *Every equilibrium isolates i if*

$$(a_i - 1/N) \inf_{0 \leq z \leq \bar{h}} A_\pi(z) \geq Da_i. \quad (36)$$

(ii) *Under strict inequality, the group rejects every incident link even with full reimbursement of its interaction cost.*

(iii) *A strict benchmark veto, $\Delta_i < 0$, persists under any admissible increasing C^1 lottery with*

$$D\|\pi - \pi_\eta\|_\infty + (Y_0 + D\bar{h})\|\pi' - \pi'_\eta\|_\infty < \frac{-\Delta_i}{a_i - 1/N}, \quad (37)$$

where the norms are on $[0, \bar{h}]$.

Proof. (i) Condition (36) makes H_i nonincreasing. Deleting a link weakly lowers innovation for every nonnegative input vector, by (5). It therefore weakly raises expected consumption and saves a positive cost.

(ii) Let $\mu = (a_i - 1/N) \inf A_\pi - Da_i > 0$. Deleting a link to obtain $\tilde{\boldsymbol{\rho}}$ raises expected consumption by at least $\mu[h(\boldsymbol{\rho}) - h(\tilde{\boldsymbol{\rho}})] > 0$, by Lemma 1. The gain remains when interaction costs are fully reimbursed.

(iii) Since $A_{\pi_\eta} \equiv \eta$,

$$\|A_\pi - \eta\|_\infty \leq D\|\pi - \pi_\eta\|_\infty + (Y_0 + D\bar{h})\|\pi' - \pi'_\eta\|_\infty.$$

The assumed bound preserves the strict inequality in (ii). □

With other expected redraw shares, the same argument replaces $a_i - 1/N$ by $a_i - \hat{a}_i$. The result establishes a robust veto. General redraw probabilities can also change the curvature of private benefits, so saturation, equilibrium existence, and the greatest-equilibrium result remain specific to the affine benchmark.

7 Conclusion

Cooperation creates resources but can threaten the shares held by powerful groups. A group may therefore refuse productive interaction even if its costs are fully reimbursed. Because links complement one another, its withdrawal can also end cooperation among groups that still value innovation positively.

The affine benchmark gives an equilibrium characterization for any number of groups and a greatest equilibrium containing all others. Lower costs and stronger productive returns expand that network. Exposure to changes in power has opposing effects across groups: full integration requires enough exposure to attract disadvantaged groups, but not so much that advantaged groups leave. These requirements define a single, possibly empty interval. The examples show that interaction can first rise and then fall as exposure increases.

The welfare loss can persist because compensating transfers require enforcement after power changes, which the benchmark excludes. Changing the distribution of future power preserves the equilibrium analysis; changing the probability rule preserves the veto under the stated marginal condition, but need not preserve the other results.

Further work could let political risk depend on who innovates, allowing groups to exclude particular partners. It could also study whether withdrawal propagates under nonlinear lotteries and which adjustment processes select among multiple equilibria.

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